

Numerical Protocol

Zumbach Effect: ESN vs. the PDV-GL Model Used as Ground-Truth DGP

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Abstract

The Zumbach-effect time-reversal asymmetry effect experiment is conducted according to the following steps:

1. **Data generation.** Use the **4-factor Markovian** Guyon & Lekeufack path-dependent volatility model (PDV-GL) [1] as the ground-truth data-generating process (DGP). The simulation employs the literature-calibrated “Table 8, Parameter Set 1 (Realistic paths)” parameters,

$$\beta_0 = 0.04, \beta_1 = -0.13, \beta_2 = 0.65, \beta_{1,2} = 0,$$

$$\lambda_{1,0} = 55, \lambda_{1,1} = 10, \theta_1 = 0.25, \lambda_{2,0} = 20, \lambda_{2,1} = 3, \theta_2 = 0.5,$$

which are subsequently rescaled to a common annualised volatility target of 20% (Section 1). This defines a single, fully specified benchmark scenario without parameter sweeps.

2. **Model calibration.** Calibrate only the ESN A2-981003 and Quadratic Rough Heston (QRH) models to the stylised-fact profile of the PDV-GL DGP. A separately calibrated copy of PDV-GL is intentionally omitted, since the DGP itself already represents the reference model.

Both models are calibrated using the complete 11-term data-adaptive score S (Eq. (4)), together with the smooth Gaussian-kernel proximity function.

3. **ESN parameterisation.** The ESN is evaluated using its current optimal architecture,

$$(N_r, N_z) = (96, 12),$$

with a read-out composed of

- a linear roughness factor;
- a per-mode z -readout profile;
- a symmetric quadratic reservoir-state interaction matrix.

These components are jointly calibrated with the Hurst exponent and the reservoir rate bounds (Section 3).

4. **Model evaluation.** Assess the ability of the calibrated models to reproduce the Zumbach time-reversal asymmetry of the DGP using
 - the lag-dependent Zumbach curve $Z(L)$;
 - the aggregate statistic

$$Z = \frac{1}{3} [Z(5) + Z(10) + Z(20)];$$

- complementary diagnostics, including the leverage autocorrelation function, volatility autocorrelation function, and return-tail statistics.

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DGP: PDV-GL as Ground Truth

The data-generating process is the Guyon & Lekeufack path-dependent volatility model [1], in its **4-factor Markovian** form – the standard finite-dimensional approximation obtained by replacing each power-law memory kernel with a mixture of two exponential (OU-type) kernels, making the model fast to simulate as a genuine Markov process:

$$\frac{dS_t}{S_t} = \sigma_t dW_t, \quad (1)$$

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}} + \beta_{1,2} R_{1,t} \sqrt{R_{2,t}}, \quad (2)$$

$$R_{1,t} = \theta_1 X_{1,0,t} + (1 - \theta_1) X_{1,1,t}, \quad R_{2,t} = \theta_2 X_{2,0,t} + (1 - \theta_2) X_{2,1,t}, \quad (3)$$

where $X_{1,j,t}$ is an exponentially-weighted moving average of past *signed* returns with mean-reversion rate $\lambda_{1,j}$ (the **leverage** factor, a mixture of a fast and a slow exponential kernel), and $X_{2,j,t}$ is an EWMA of past *squared* returns with rate $\lambda_{2,j}$ (the **magnitude** factor). Unlike every other DGP in this line of protocols, PDV-GL is used here as **ground truth**, not as a candidate model.

”True” parameters – literature-calibrated Table 8, Parameter Set 1 (Realistic paths).

λ rates are in annualised units; each is applied at the daily step via $e^{-\lambda \Delta t}$, $\Delta t = 1/252$.

Rescaling. Run literally with these parameters (and daily returns in decimal units), the model’s realised annualised volatility comes out at only $\approx 4\%$ – the source table’s implicit return-scale convention could not be verified from the parameters alone. A single, documented linear rescaling $X \mapsto X \cdot (\text{TV_DAY}/\hat{\sigma})$ is applied post-hoc to bring the level to the same 20% annualised target used throughout this line of protocols. This is a pure amplitude change: every *normalised* statistic used in this protocol ($\hat{H}$, kurtosis, leverage correlation, and the $Z(\tau)/\sigma^3$ cross-moment ratio) is left **exactly unchanged**, since numerator and denominator scale identically under a uniform linear rescaling.

Zumbach effect mechanism. Because $\beta_1 < 0$, a large *negative* past return raises $R_{1,t}$ with the sign needed to raise σ_t and hence future variance – a genuine, signed leverage asymmetry. At these literature parameters, the realised leverage correlation is ≈ -0.04 to -0.17 depending on

the exact path sample, and the windowed Zumbach scalar Z (Eq. 4’s calibration statistic) comes out strongly positive ($Z \approx 0.22$ in the run reported here) – a clearly non-trivial, sustained effect rather than a borderline one.

Simulation. Eq. (3) is iterated as an exact discretisation of the four OU-type factors $(X_{1,0}, X_{1,1}, X_{2,0}, X_{2,1})$ at the daily step, using $e^{-\lambda_{i,j}\Delta t}$ decay – a genuine Markov process in $(X_{1,0}, X_{1,1}, X_{2,0}, X_{2,1})$, no path history storage needed (unlike the power-law-kernel DGPs used elsewhere in this line of protocols).

Parameter grid. None – Parameter Set 1 is a single, fully-specified literature calibration (Table 1); every path uses the same fixed parameters.

Models

2.1 ESN A2-981003 (current optimal architecture)

The ESN uses Gaussian Brownian innovations, with 7 shared hyperparameters and $N_r = 96, N_z = 12$. **rough_scale** and the z -readout (**zr_lo**, **zr_hi**, plus the symmetric quadratic matrix coefficients (m_1, m_2)) are calibrated per DGP cell (§3), not part of this fixed table. Single $\varepsilon_t \sim \mathcal{N}(0, 1)$ drives both the Ito log-return and the diagonal OU rough bank. The negative instantaneous leverage $\kappa_0 = -0.434991 < 0$ (admissibility, guaranteed by construction) gives the ESN an endogenous Zumbach-like asymmetry – structurally different from PDV-GL’s own path-dependent-feedback mechanism (§1), so a good match here is evidence the ESN’s mechanism can *mimic* the effect, not that it replicates the same causal channel. H , the r -layer rate window $[\lambda_{lo}, \lambda_{hi}]$, and the z -layer rate window $[a_{lo}, a_{hi}]$ are **calibrated per α_r DGP cell** (§3) – they are not part of the shared architecture above.

2.2 Quadratic Rough Heston (QRH)

Bourgey & Gatheral: $V_t = Y_t^2 + c$, gamma kernel $\kappa(\tau) = \nu/\Gamma(\alpha)\tau^{\alpha-1}e^{-\lambda\tau}$, $\alpha = H + \frac{1}{2}$, Euler-Volterra with 252-step ring buffer. Included alongside the ESN as a second, structurally different candidate model (a genuine rough-volatility SDE, vs. the ESN’s Markovian reservoir approximation) against the same PDV-GL ground truth.

Calibration: Full 11-Term Score, Smooth Kernel

The data-adaptive score (identical structure to every other protocol in this series, Eq. (4)), with the **smooth Gaussian-kernel** proximity (protocol V5) rather than the original piecewise-linear tent:

$$\begin{aligned} S = & 2.2 (1 - f(\hat{H})) + 1.4 (1 - f(\bar{\sigma})) + 0.6 (1 - g(q_{995})) + 0.2 (1 - g(\sigma_{\max})) \\ & + 0.8 (1 - f(\bar{V}_{ACF})) + 1.0 (1 - f(G_T)) + 0.8 (1 - f(F_T)) + \mathbf{3.5} (1 - f(Z)) \\ & + 1.1 (1 - f(L)) + 0.7 (1 - f(K)) + 0.5 (1 - f(A)) + 5 \cdot \text{Stress}, \end{aligned} \quad (4)$$

where $f(x, c, s) = \exp(-\frac{1}{2}((x - c)/s)^2)$, $g(x, c, s) = \exp(-\frac{1}{2}((x - c)_+/s)^2)$, both in $(0, 1]$. $S \geq 0$ is **minimised directly** (no sign flip); $S = 0$ is a perfect match on all 11 stylised facts and no stress event; $S = 17.8$ is **worst-possible** (raised from 15.3 by the Zumbach-weight increase below). All

Table 1: PDV-GL 4-factor Markovian ground-truth parameters (Table 8, Parameter Set 1).

Parameter	Value
β_0	0.04
β_1	-0.13
β_2	0.65
$\beta_{1,2}$	0 (unused for Set 1)
$\lambda_{1,0}$	55
$\lambda_{1,1}$	10
θ_1	0.25
$\lambda_{2,0}$	20
$\lambda_{2,1}$	3
θ_2	0.5

Table 2: The 7 shared hyperparameters of the optimal architecture.

Hyperparameter	Value
N_r	96
N_z	12
z_strength	0.1929
even_strength	2.2291
linear_strength	0.1930
gamma_norm	0.9651
local_z_strength	0.0476
zz_scale	0.0397
sign_prob_neg	0.2219
rough_orientation	-1.0 (fixed)

centres c_k and tolerances s_k are DGP cross-path mean and $\max(\text{std}, 0.30|c_k|, \text{floor}_k)$.

Zumbach score weighting. The Zumbach weight is raised from 1.0 to **3.5** – now the single largest weight in S , exceeding even $\hat{H}$ ’s 2.2 – and QRH’s calibration uses a 2-point multi-start. §?? reports the full root-cause diagnosis and result.

ESN read-out. The ESN’s read-out combines a linear rough factor, a per-mode z -readout profile (`zr_lo`, `zr_hi`), and a symmetric quadratic-term matrix $Q = (1/N_r)(m_1 I + m_2 q q^\top)$ added to the pre-activation. The ESN calibrates its full inner-loop vector

$$H, \lambda_{\text{lo}}, \lambda_{\text{hi}}, a_{\text{lo}}, a_{\text{hi}}, \Delta b_0, \text{scale}, \text{rough_scale}, \text{zr_lo}, \text{zr_hi}, m_1, m_2$$

– **12 dimensions** – via an 11-point multi-start Nelder–Mead scheme (rough/fast, persistent/slow, intermediate, wide-band/multiscale, long-memory extreme, Zumbach term1/term2-balanced, strong-amplitude, weak-amplitude, isotropic-positive, isotropic-negative, and q -aligned-positive starts), keeping the best-of-eleven result; only the 7 shared architecture hyperparameters of Table 2 stay fixed.

Calibrated parameters:

Model	Parameters calibrated	Remark
ESN A2-981003	$H, \lambda_{\text{lo}}, \lambda_{\text{hi}}, a_{\text{lo}}, a_{\text{hi}}, \Delta b_0, \text{scale } s, \text{rough_scale}, \text{zr_lo}, \text{zr_hi}, m_1, m_2$	7 shared hyperparameters fixed; full 12-dim. inner-loop calibrated per α_r cell (11-start Nelder–Mead)
QRH	$H, \nu_{\text{vol}}, \lambda, c_{\text{frac}}$	2-start Nelder–Mead (added a low- H start; see §??)

Zumbach Diagnostic

$$Z(L) = \text{Corr}(\text{past}_L R^2, \text{fut}_L V) - \text{Corr}(\text{past}_L V, \text{fut}_L R^2), \quad (5)$$

where $\text{past}_L R^2 = (X_t - X_{t-L})^2$ and $\text{fut}_L V = (V_{t+L} - V_t)/L$. $Z(L) > 0$ means the past-magnitude→future-vol channel $[\text{Corr}(\text{past}_L R^2, \text{fut}_L V)]$ dominates the reverse, time-reversed channel $[\text{Corr}(\text{past}_L V, \text{fut}_L R^2)]$ – the genuine Zumbach / time-reversal-asymmetry signature. This is the SAME windowed, correlation-based metric used for calibration (the scalar $Z = \frac{1}{3}[Z(5)+Z(10)+Z(20)]$ enters the score with weight 1.0) and for Figure Z1. It depends only on magnitudes, not the sign of returns, and so is well-defined even for DGPs with no leverage term at all.

The ESN carries its own, structurally distinct source of asymmetry (the fixed $\kappa_0 < 0$ leverage built into its architecture, §2) and the PDV-GL DGP its own path-dependent feedback (§1) – a good match between the ESN’s and the DGP’s $Z(L)$ is therefore evidence that the ESN’s mechanism can *mimic* the magnitude of the effect, not that it reproduces the same causal channel.

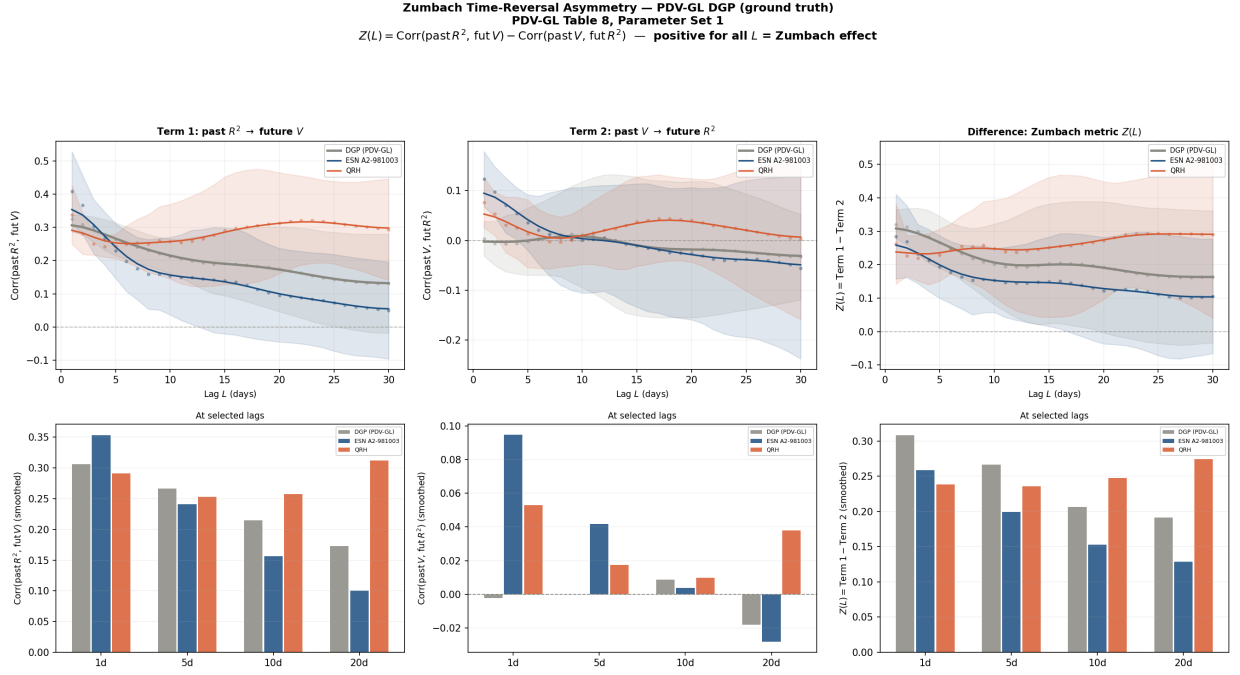


Figure 1: Zumbach windowed metric, decomposed into its two terms. Column 1: $\text{Corr}(\text{past } R^2, \text{fut } V)$ (the genuine, forward Zumbach channel); column 2: $\text{Corr}(\text{past } V, \text{fut } R^2)$ (the reverse, time-reversed channel); column 3: $Z(L) = \text{column 1} - \text{column 2}$ (the Zumbach metric itself). Top row: curve vs. lag (Gaussian-smoothed mean ± 1 std band); bottom row: grouped bars at lags 1d, 5d, 10d, 20d. Showing the two terms side by side makes it possible to see whether a model matches the DGP’s $Z(L)$ because both terms individually match, or because two mismatched terms happen to cancel to a similar difference.

Zumbach Effect $Z(L) = \text{Corr}(\text{past } R^2, \text{fut } V) - \text{Corr}(\text{past } V, \text{fut } R^2)$
PDV-GL DGP (ground truth) | $Z(L) > 0$ = Zumbach effect present

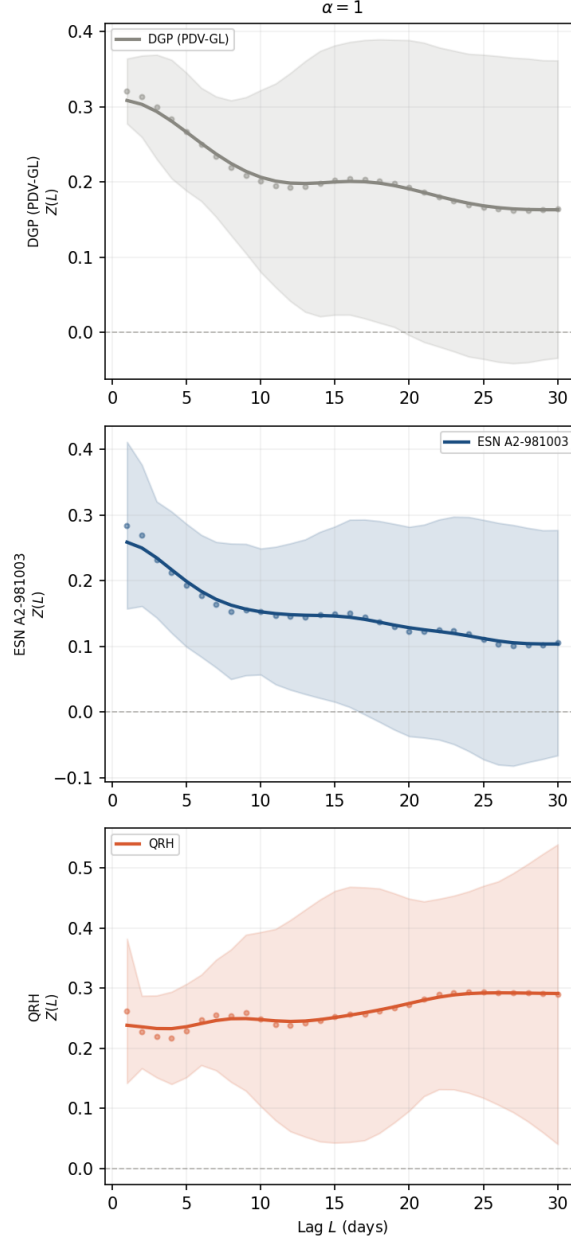


Figure 2: Windowed $Z(L)$ curves, one row per source (DGP, ESN, QRH), for the single α_r cell tested. Dots are the raw per-lag statistic; the line is Gaussian-smoothed ($\sigma_{\text{sm}} = 2$ days); the shaded band is ± 1 std across paths.

Figures

Main figure (Zumbach windowed metric), reproduced above as Figure 1, and Figure 2 (Figure Z1) are placed in §4, alongside the metric they illustrate.

Monte Carlo Setup

Parameter	Value
DGP paths n_{dgp}	30
Calibration paths n_{cal}	5
Calibration days T_{cal}	600
Calibration burn-in	150
Final paths n_{sim}	30
Final days T_{sim}	2500
Final burn-in	500
Δt	1 (daily)
DGP	PDV-GL Table 8, Parameter Set 1 (fixed, no grid)
DGP $\beta_0, \beta_1, \beta_2$	0.04, -0.13 , 0.65
DGP $\lambda_{1,0}, \lambda_{1,1}, \theta_1$	55, 10, 0.25
DGP $\lambda_{2,0}, \lambda_{2,1}, \theta_2$	20, 3, 0.5
Smoothing σ	2 days

References

- [1] J. Guyon, J. Lekeufack. *Volatility is (mostly) path-dependent*. QF 23(9), 2023.
- [2] G. Zumbach. *Time reversal invariance in finance*. Quantitative Finance 9(5):505–515, 2010.

DGP + Model Diagnostics — PDV-GL (ground truth)
Histogram | Hill | Leverage ACF | Volatility ACF

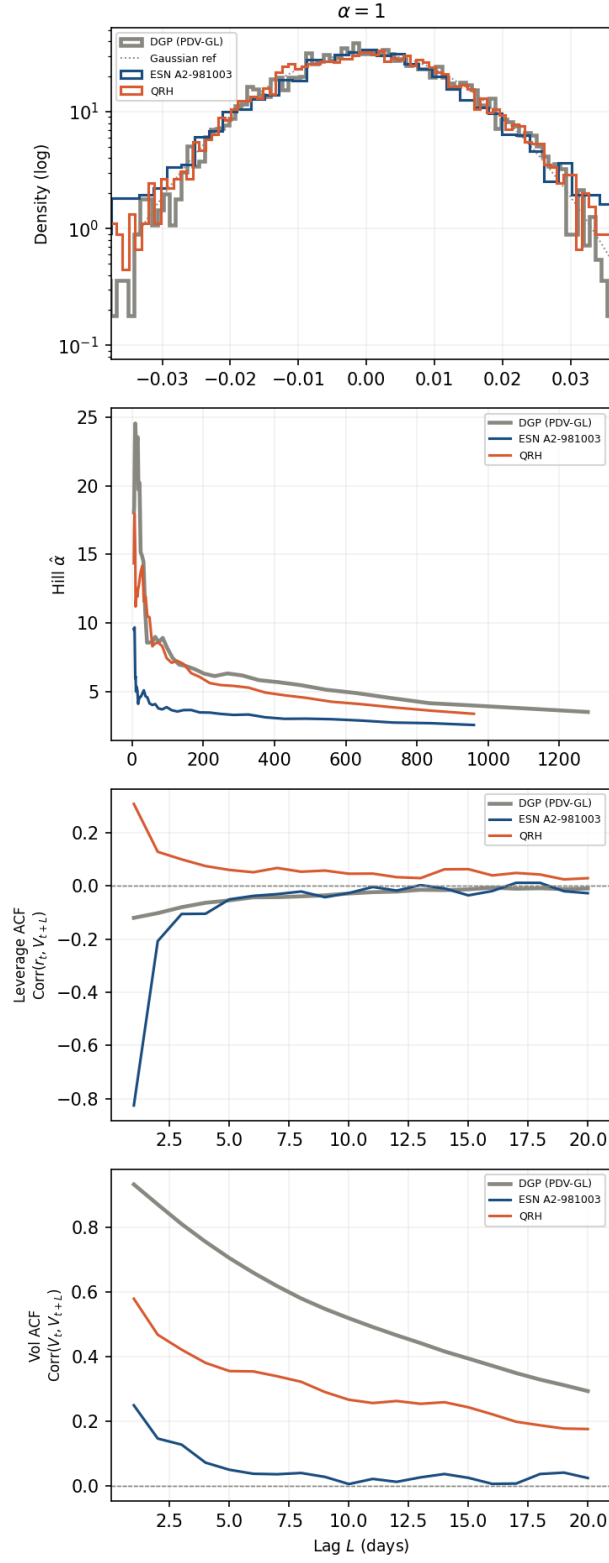


Figure 3: Return and volatility diagnostics, DGP vs. ESN vs. QRH: log-density histogram vs. Gaussian reference (top); Hill estimator $\hat{\alpha}$ (second); leverage ACF $\text{Corr}(r_t, V_{t+L})$ (third); volatility ACF $\text{Corr}(V_t, V_{t+L})$ (bottom).